



Astronaut to vote from space

NASA astronaut Kate Rubins will vote in the upcoming presidential election from space. <http://digit.in/oct20-39>



New water bodies found on Mars

Using data from Mars Express, researchers have found three new reservoirs of liquid water on Mars. <http://digit.in/oct20-40>

There is no such thing as a fish

What even is a fish?

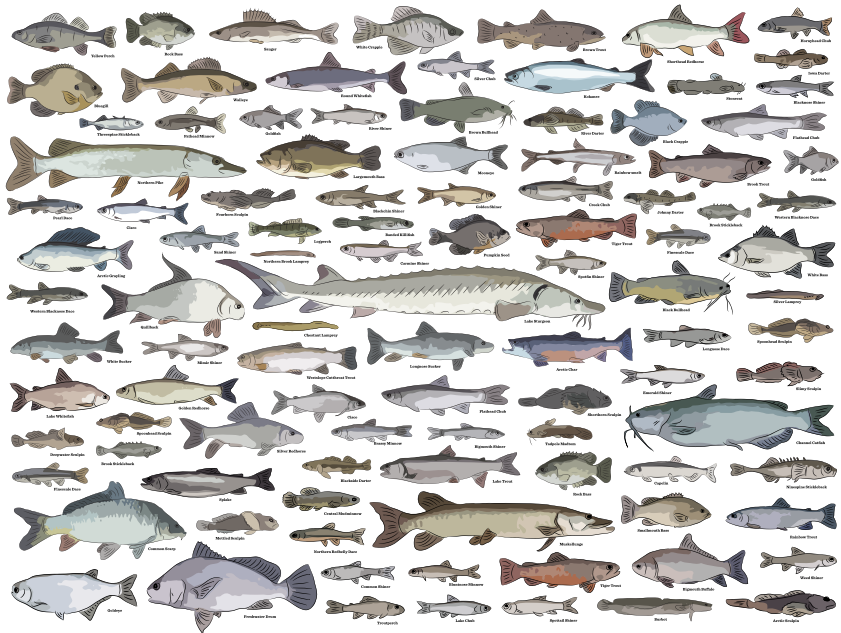
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he taxonomic ranks, from the lowest to the highest is species, genus, family, order, class, phylum,

kingdom and domain. The Bombay Duck for example is scientifically known as *Harpadon nehereus*, belongs to the genus of *Harpadon*, the family of *Synodontidae*, the order *Aulopiformes*, the class *Actinopterygii*, the phylum *Chordata*, the kingdom *animalia* and the domain *Eukarya*. At every taxonomic rank or level, there was a single common ancestor. Sure, all life on Earth had the same common ancestor, but that is why we have so many ranks in the taxonomic system, to keep track of increasingly more recent common ancestors. The higher the taxonomic rank, the older the divergence for that rank occurred. For example, all the species that make up the genus *Harpadon*, could not have diverged before all the species that make up *Chordata*. Now, scientists can look at the DNA and life on a molecular level to classify it, but for most of history we have been simply looking at the morphology of the animals to figure out their relationships. As horses, zebras and donkeys look alike, we classify all of them under the genus of *Equus*, it is as simple as that.

Now there are a wide variety of creatures that are classified under the phylum of *Chordata*, and most of them belong to a subphylum called



Vertebrata, which are chordates with backbones. Vertebrata have been divided into 7 classes, which are jawless fishes, cartilaginous fishes, bony fishes, amphibians, reptiles, birds and mammals. Scientifically, these classes are *Agnatha*, *Chondrichthyes*, *Osteichthyes*, *Amphibia*, *Reptilia*, *Aves* and *Mammalia*. Apart from these, there are two extinct classes, *Placodermi* and the *Acanthodii*, both of which consisted of armoured fishes. Since the Cambrian explosion, over hundreds of millions of years, we saw the divergence of the classes. The mammals split up from the reptiles long before the birds diverged from them. All the three of these classes, reptiles, birds and mammals diverged from amphibians. All amphibians, and by extension reptiles, birds and mammals diverged from bony fishes. So far, so good, everything is neat and tidy. Bony fish, amphibians, reptiles,

birds and mammals all had the same common ancestor, around 420 million years ago.

The problem is, jawless fishes, cartilaginous fishes and bony fishes are all considered to be fishes. *Agnatha*, *Chondrichthyes*, and *Osteichthyes* are all considered fishes, but four classes of animals diverged from *Osteichthyes*. Salmon, catfish, sharks, rays, lampreys and hagfish are all considered fish, but the common ancestor for all these diverse species is also the common ancestor for donkeys, horses, ducks, salamanders, pigeons and crabs. Biologically, the very idea of a fish is deeply flawed and does not make sense. In the biological sense, a fish is defined by exclusion of other vertebrates. There is no single branch in the evolutionary tree, that you can identify and say, that all the descendants of this organism is a fish, because that would include some things that